AARTI STEEL INTERNATIONAL LIMITED
Ludhiana, Punjab

INDUSTRIAL INTERNSHIP REPORT

MANUFACTURING PROCESS & QUALITY TESTING OF ROLLED STEEL BARS

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1. Introduction

This report describes the complete manufacturing process employed for the production of hot-rolled steel bars, covering the full sequence of operations from receipt of raw steel billets through to final inspection, testing, packing, and dispatch of finished bar products. The document is intended as a technical reference for production, quality assurance, and training purposes, and outlines both the metallurgical rationale and the practical shop-floor sequence followed at a typical integrated bar rolling mill.

1.1 Hot Rolling Process

Hot rolling is a metal-forming process carried out above the recrystallisation temperature of steel, typically between 1050°C and 1250°C. At these temperatures the steel is significantly more ductile and offers lower resistance to deformation, allowing large reductions in cross-sectional area to be achieved with relatively modest rolling loads. As the billet passes through successive roll stands, its cross-section is progressively reduced and elongated, while its temperature gradually falls due to radiative, convective, and contact heat losses, as well as the heat consumed in plastic deformation.

The hot rolling route offers several advantages over cold working for long products of this size: it refines the coarse as-cast grain structure inherited from continuous casting, closes internal porosity through the compressive deformation of the rolling bite, and permits large section reductions in a short time without intermediate annealing. The finishing temperature of rolling is carefully controlled since it directly influences the final grain size, mechanical properties, and, in the case of TMT bars, the effectiveness of the subsequent quenching and self-tempering treatment.

1.2 Finished Product Range

The rolling mill produces a diverse range of steel sections serving construction, infrastructure, automotive, and industrial markets. Typical products are summarised below.

Product Type	Description	Typical Application
TMT Bars	Thermo-Mechanically Treated rebars	Reinforced concrete structures
Wire Rods	Coiled fine-section rounds	Wire drawing, fasteners, springs
Round Bars	Solid circular cross-sections	Shafts, axles, machined components
Flat Bars	Rectangular cross-section flats	Fabrication, general engineering
Square Bars	Equal square cross-sections	Machined parts, ornamental work
Structural Sections	Angles, channels, beams	Construction frameworks

1.3 Process Flow Overview

The rolling mill process follows a well-defined sequence of interdependent stages. The table below summarises the key stages, their primary function, and the critical process parameter monitored at each.

Stage	Function	Key Parameter
Billet Production (CCM)	Solidification of molten steel into billets	Mould cooling, withdrawal speed
Billet Storage	Grade-wise storage and traceability	Heat number, grade marking
Billet Charging	Controlled feeding into furnace	Charging rate, spacing
Reheating Furnace	Uniform heating to rolling temperature	1100–1250°C discharge temp
Descaling	Removal of surface oxide scale	150–250 bar water pressure
Roughing Mill	Initial cross-section reduction	High compressive force, 90° rotation
Intermediate Mill	Progressive dimensional refinement	Speed synchronisation
Finishing Mill	Final dimensions and surface finish	850–1000°C finishing temp
Cooling Bed	Controlled air cooling	Residence time by product size
Straightening	Dimensional correction post-cooling	Multi-roller pressure
Cold Shearing	Cutting to commercial lengths	6 m / 9 m / 12 m lengths
Bundling & Dispatch	Identification, packaging, and delivery	Bundle weight, grade tag

This is followed downstream by the quality assurance testing chain — PMI, MPI, Auto UT, Macro Examination, and Laboratory Testing — described in detail in Section 10, before final release, packing, and dispatch.

Figure 1: **Rolling Mill Workflow**

2. Billet Production and Receiving

2.1 Billet Production by Continuous Casting

The rolling process originates at the Continuous Casting Machine (CCM). Molten steel, refined in the Steel Melting Shop (SMS), is transferred via ladle to a tundish, an intermediate buffer vessel that maintains a steady, controlled flow of liquid steel into the water-cooled copper mould.

Inside the mould, rapid heat extraction causes a solid shell to form around the molten core. As the strand is withdrawn downward by pinch rolls, secondary spray cooling completes solidification through the strand's cross-section. The solidified billet is then straightened and continuously cut to predetermined lengths using hydraulic or oxy-fuel shears.

Each billet is marked with heat number, steel grade, and identification details prior to transfer to the billet storage yard, ensuring full traceability from casting to finished product.

2.2 Source of Billets

Billets may be received from the plant's own steel melting shop and continuous casting machine (a fully integrated operation), or procured from external suppliers as merchant billets. In either case, each batch is accompanied by a mill test certificate (MTC) or heat-wise chemistry report confirming the grade, heat number, and chemical composition of the steel.

2.3 Billet Specifications

Typical billet cross-sections used for bar rolling range from 100 mm × 100 mm to 150 mm × 150 mm square, in standard lengths of 6 to 12 metres, depending on the mill's capacity and the final product size being rolled. Key specification parameters include chemical composition (grade), cross-sectional size and squareness tolerance, straightness, surface cleanliness, and absence of internal defects such as porosity or pipe.

2.4 Billet Storage

Billets are stacked heat-wise and grade-wise in a designated storage yard, with each stack clearly tagged to preserve traceability back to the casting heat — the same heat number and grade marking applied at the CCM. Segregation by grade prevents mix-ups during charging into the reheating furnace, and stacking height and orientation are controlled to avoid bending or surface damage.

2.5 Billet Inspection

Before charging, billets are visually inspected for surface defects such as cracks, cold shuts, rhomboidity (off-square cross-section), scale pits, and residual mould flux entrapment from the casting process. Dimensional checks confirm cross-section size and length, and any billet showing severe surface or internal defects is set aside for conditioning (grinding) or rejection prior to entering the furnace.

3. Reheating Furnace

Figure 2: **Reheating Furnace**

3.1 Purpose of Reheating

The reheating furnace is a critical unit that brings the billet to the correct rolling temperature in a uniform and energy-efficient manner. Its purpose is to raise the billet temperature uniformly to the range required for hot rolling typically 1100°C to 1250°C at the furnace discharge end so that the steel is sufficiently plastic to be deformed with the rolling loads available at the mill, while ensuring a homogeneous temperature (and hence homogeneous microstructure and flow behaviour) through the full cross-section of the billet.

3.2 Furnace Construction and Mechanism

Modern steel plants employ pusher-type, walking beam, or walking hearth furnaces fired by natural gas, furnace oil, or mixed gas fuel systems, fitted with recuperators to preheat combustion air using waste flue gas heat. Billets are advanced through the furnace using either a walking beam mechanism — in which a movable beam lifts, advances, and lowers the billets in a stepwise cycle — or a walking hearth/pusher-type arrangement, depending on the specific furnace design installed at the plant. The walking beam design is generally preferred in modern mills as it minimises skid marks (localised cold spots) on the billet underside and allows more flexible control of billet residence time.

3.3 Heating Zones

The furnace is divided into three thermal zones, each performing a distinct role in bringing the billet from ambient to rolling temperature.

(a) Preheating Zone

Incoming billets absorb waste heat from hotter furnace sections, gradually rising from ambient to 500–800°C. This gradual heating prevents thermal shock, minimises internal stress cracking, and improves overall furnace thermal efficiency.

(b) Heating Zone

High-intensity burners rapidly increase billet temperature toward rolling target levels, reaching 900–1150°C. Heat is transferred via radiation from flames, convection of combustion gases, and limited internal conduction. The billet surface transitions from bright orange to yellow as temperature rises.

(c) Soaking Zone

The soaking zone eliminates temperature gradients between the billet surface and core, achieving uniform temperature at 1100–1250°C. Adequate soaking reduces rolling loads, ensures homogeneous deformation, and improves product dimensional consistency.

Furnace Zone	Temperature Range	Primary Function
Preheating Zone	500 – 800°C	Gradual initial heating; waste heat recovery
Heating Zone	900 – 1150°C	Rapid temperature increase to near-rolling temp
Soaking Zone	1100 – 1250°C	Surface-to-core temperature equalisation

3.4 Temperature Control

Zone temperatures are monitored using thermocouples and controlled through automated combustion control systems that regulate fuel and air flow to each zone. Furnace discharge temperature is verified using pyrometers, and heating rate and total residence time are set according to billet size and steel grade to avoid overheating (which can cause grain coarsening or incipient melting at grain boundaries) or underheating (which increases rolling loads and can lead to surface tearing).

3.5 Scale Formation

During reheating, the billet surface oxidises in the furnace atmosphere, forming iron oxide scale — a layered structure of FeO, Fe_3O_4, and Fe_2O_3. Excessive scale formation causes material yield loss (typically 1–2% of billet weight under normal control), surface defects, increased roll wear, and poor dimensional control. Furnace atmosphere composition and residence time are therefore carefully controlled to minimise this loss.

3.6 Billet Discharging

Once soaked to the target temperature, billets are discharged from the furnace through a discharge door onto the roller table leading to the descaling station, in a sequence synchronised with the rolling mill's production rate to maintain continuous, uninterrupted rolling.

4. Oxidation and Descaling

Figure 3: **De-Scaler**

4.1 Need for Descaling

The oxide scale formed on the billet surface during reheating is harder and less ductile than the underlying steel. If not removed before rolling, this scale is pressed into the bar surface by the rolling force, producing surface pitting and poor surface finish, and can also accelerate wear of the roll grooves. To mitigate this, furnace atmosphere composition and residence time are carefully controlled, and descaling is carried out immediately after furnace discharge and before the billet enters the first (roughing) stand.

4.2 High-Pressure Water Descaling

Upon exit from the furnace, billets pass through a high-pressure water spray descaling system operating at 150–250 bar, arranged around the billet as it passes through the descaling box. The high-impact water jets thermally shock and mechanically fracture and remove the brittle scale layer from the hot billet surface without significantly cooling the billet core.

4.3 Benefits of Effective Descaling

- Improved surface finish and product appearance
- Enhanced dimensional accuracy and tolerance compliance
- Reduced roll wear and extended roll life
- Prevention of surface-embedded scale defects

5. Rolling Mill Configuration

The rolling mill itself consists of a series of roll stands arranged in sequence, each fitted with grooved rolls that progressively reduce the billet cross-section and shape it toward the final bar profile. The stands are grouped into roughing, intermediate, and finishing mill sections.

5.1 Roughing Mill

The roughing mill performs the primary deformation stage and is the first group of stands that the descaled billet enters. Large-diameter rolls apply substantial compressive forces to dramatically reduce the billet's cross-sectional area while significantly increasing its length. The billet is rotated 90° between passes using twisting guides to ensure uniform deformation and grain refinement in all orientations. This stage is responsible for breakdown of the coarse as-cast grain structure, promotion of dynamic recrystallisation for grain refinement, and uniform material flow preparation for intermediate rolling.

5.2 Intermediate Mill

The intermediate mill performs progressive dimensional reduction while improving shape accuracy. Multiple stands are arranged in horizontal, vertical, or alternating configurations. Guide systems ensure precise billet alignment before each pass, preventing twisting and misalignment. Rolling speed increases progressively to compensate for elongating bar length while temperature remains above the recrystallisation threshold.

5.3 Finishing Mill

The finishing mill produces the final product geometry and surface quality, including, for ribbed bars, the transverse and longitudinal rib pattern rolled into the surface by the finishing groove design. Closely spaced stands operating at high speed apply small, controlled reductions with precise roll gap and speed management. The finishing temperature is maintained at 850–1000°C to ensure appropriate microstructural development and surface quality — this is a critical process parameter since it governs final grain size and, for TMT bars, the effectiveness of the downstream quenching and self-tempering treatment.

Rolling Stage	Roll Diameter	Reduction Level	Primary Purpose
Roughing Mill	Large	High (major)	Cross-section breakdown, grain refinement
Intermediate Mill	Medium	Moderate	Progressive dimensional refinement
Finishing Mill	Small	Low (precise)	Final geometry, surface finish

5.4 Reduction in Cross-Section and Shape Transformation

Across the full stand sequence, the billet cross-section is reduced from, for example, 130 mm × 130 mm square down to a finished bar diameter of 8–32 mm, representing a very large total reduction ratio achieved over 16 to 20 (or more) individual passes, depending on mill layout and final bar size. The bar cross-section evolves through a defined sequence of pass shapes with each pass designed to progressively reduce area while controlling metal spread and avoiding fold-over defects (laps) at the shoulders of the section.

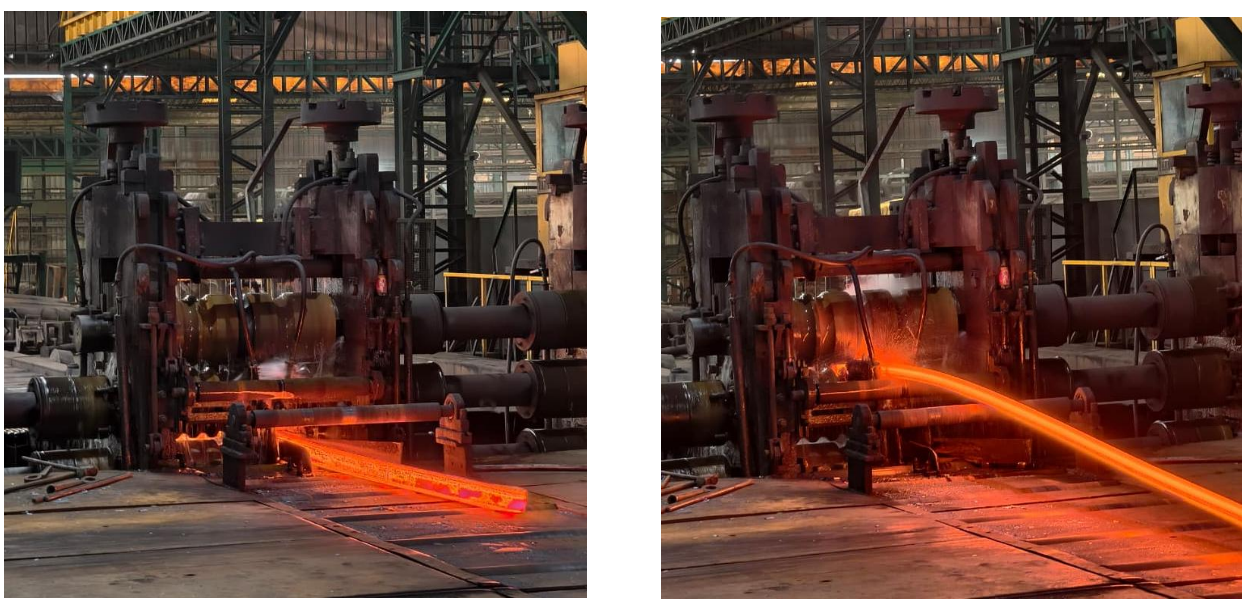

Figure 4: **Rolling Mill & Size Reduction**

5.5 Roll Pass Design

Roll pass design is the engineering of the groove sequence machined into the rolls through which the steel progressively passes. Each pass is designed to achieve controlled metal flow, reduce rolling load, and minimise internal stresses. A well-designed pass schedule is fundamental to product quality, roll life, and mill productivity.

Advantages of a properly engineered pass schedule include:

- Minimises rolling load and energy consumption per tonne
- Prevents surface defects such as seams, laps, and fins
- Reduces roll groove wear and extends campaign life
- Improves dimensional accuracy and product tolerances
- Maximises mill throughput and productivity

5.6 Dimensional Control and Temperature During Rolling

Bar dimensions are continuously monitored, historically by manual micrometer/caliper checks and increasingly by inline laser or optical gauges that provide real-time diameter feedback to the mill's automatic gauge control system, allowing roll gap and speed adjustments to hold dimensional tolerances within specification. Bar temperature is tracked through the mill using pyrometers at key locations (after roughing, after intermediate, and at finishing mill entry/exit).

6. Finishing and Post-Rolling Operations

6.1 Cooling Bed

After the final rolling pass (and, for TMT bars, after passing through an inline quenching box), the product is transferred at high speed onto a racking cooling bed. Stationary and reciprocating racks distribute bars uniformly across the bed, where they cool by natural air convection. Controlled cooling relieves residual thermal stresses, prevents distortion, and ensures temperature uniformity along the bar length before it proceeds to cutting operations. Cooling rate on the bed also influences the final microstructure and mechanical properties of the bar, particularly for TMT bars, where the self-tempering of the martensitic outer rim by residual core heat takes place during this stage.

6.2 Straightening

Differential thermal contraction during cooling can produce slight bowing or camber in rolled bars. Multi-roller straightening machines apply controlled bending forces to restore required straightness. Straightening ensures dimensional compliance with applicable standards and facilitates downstream handling and bundling.

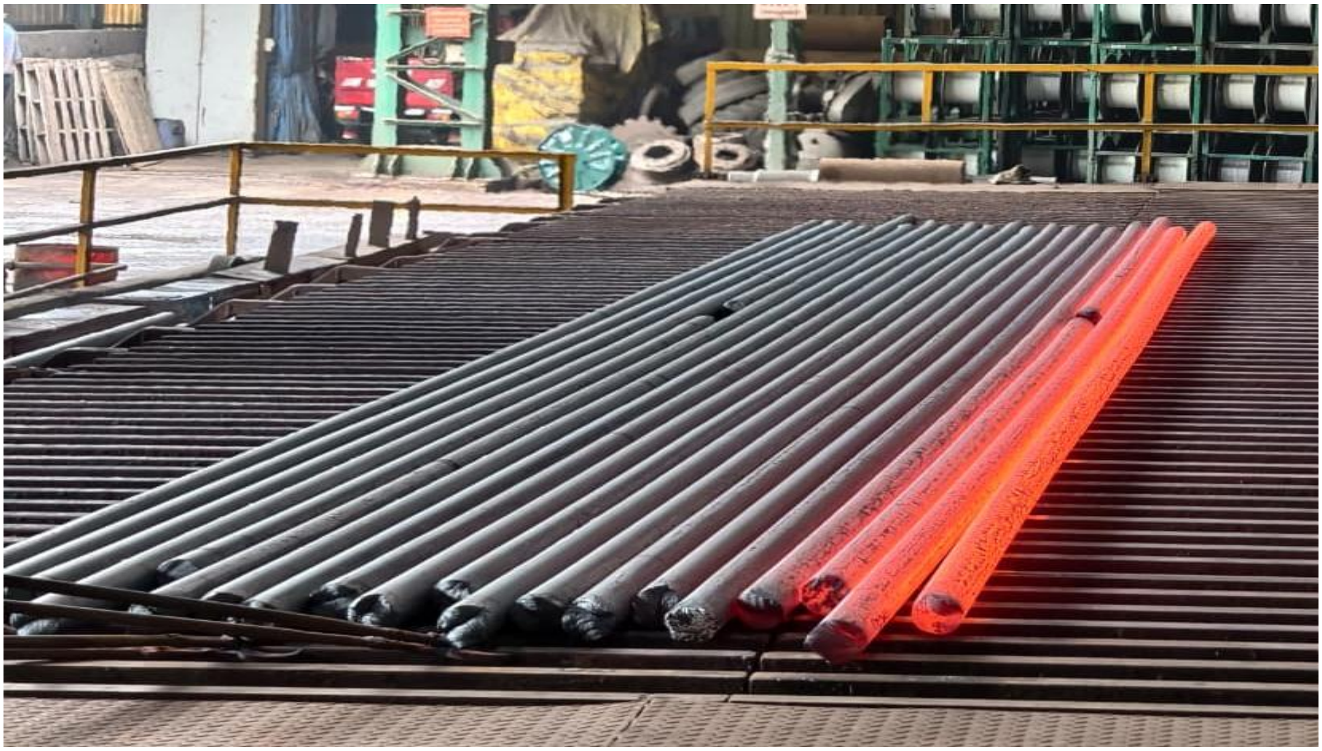

Figure 5: **Cooling Bed**

Figure 6: **straightening machines**

6.3 Cold Shearing

Cooled bars are cut to standard commercial lengths using hydraulic or mechanical cold shears. Standard lengths of 6 m, 9 m, and 12 m are produced, with custom lengths available to customer specification. Cold shearing produces clean, square-cut ends with minimal deformation or burring. Cutting length is verified against a stop-gauge or measuring wheel to maintain length tolerance.

6.4 Counting, Bundling, and Identification

Cut bars are automatically counted and assembled into bundles of predetermined weight or quantity. Steel strapping secures each bundle for handling and transport. Each bundle receives an identification tag carrying the steel grade, product size, heat number, batch number, production date, and bundle weight — enabling complete traceability through the supply chain.

Figure 7: **Stacking Area**

7. Surface Preparation - Shot Blasting

7.1 Shot Blasting

Where specified, bundled or individual bars are passed through a shot blasting chamber in which small steel shot is propelled at high velocity onto the bar surface by rotating blast wheels, mechanically removing any residual scale, rust bloom, or surface contamination.

7.2 Purpose of Shot Blasting

Shot blasting improves surface cleanliness ahead of inspection and downstream coating or galvanising operations, and also assists in revealing surface defects (such as fine seams or laps) that might otherwise remain masked under a layer of scale, thereby supporting more reliable visual and NDT inspection.

7.3 Surface Finish after Shot Blasting

After shot blasting, the bar surface presents a uniform, matte-grey metallic finish free of loose scale, which provides a consistent background for subsequent visual inspection and improves the reliability of magnetic particle inspection by ensuring good contact and even distribution of inspection medium.

Figure 8: **Shot Blast**

8. Visual and Dimensional Inspection

8.1 Visual Inspection

Trained inspectors examine the full length of each bar (or a representative sample per bundle, per applicable sampling standard) under good lighting conditions to identify visible surface defects, checking systematically along the bar length and around its circumference.

8.2 Dimensional Inspection

Bar diameter, rib height and spacing (for ribbed bars), ovality, and length are checked against the applicable product standard using calibrated measuring instruments (micrometres, vernier callipers, rib gauges, and measuring tapes/wheels), with results recorded against heat number for traceability.

8.3 Surface Defect Identification

Surface defects identified during visual inspection are classified by type, size, and location, and disposed of according to the applicable acceptance standard — accepted as-is if within tolerance, referred for grinding/reconditioning, or rejected if defects exceed permissible limits. The principal surface and near-surface defect types encountered in rolled bar production, their characteristic appearance, and their typical origins are described in detail below.

(a) Lapping (Laps)

A lap is a surface defect formed when a thin tongue or fin of metal, produced by excess metal filling a roll groove (often due to oversized incoming stock, roll groove mismatch, or worn/misaligned roll passes), is folded back onto the bar surface during a subsequent rolling pass rather than being fully worked into the body of the bar. Because the fold-over surfaces do not weld together under rolling pressure — they are already oxidised — a lap appears as a fine, often intermittent line running roughly parallel to the rolling direction, sometimes with a slightly raised or feathered edge. Laps are a serious defect because they create a discontinuity that reduces effective cross-sectional area and can act as a stress-riser and crack initiation site under load. They are typically detected by visual inspection and confirmed by magnetic particle inspection, and are addressed by adjusting groove design/roll gap settings and by grinding out isolated occurrences.

(b) Sliver

A sliver is a thin, elongated sliver-like flake of steel that remains partially attached to the bar surface, oriented along the rolling direction. Slivers typically originate either from surface defects already present on the incoming billet (such as sub-surface inclusions, blowholes, or cast surface laps that are rolled out) or from metal that is torn loose from a lap or fold during rolling and dragged along the surface in a subsequent pass. Slivers appear as narrow, tongue-like projections that may be loosely attached at one end, and can be lifted with a blade or fingernail during inspection. Left in place, slivers can break away in service or during subsequent processing, leaving a shallow surface depression, and they are removed by local grinding once identified.

(c) Seams

A seam is a fine, longitudinal surface discontinuity running along the length of the bar, generally straight and narrow, that originates from an as-cast surface defect on the billet — commonly a longitudinal crack, cold shut, or pinhole opened up on the billet surface during casting or cooling — which is then elongated and drawn out along the bar axis as the billet is rolled down through successive passes. Unlike a lap, a seam does not necessarily involve folded-over metal; it is more often a closed or partially closed crack-like line at the surface. Seams can extend to a shallow depth below the surface and, if deep or numerous, can significantly affect fatigue performance and bend/weld quality. They are identified visually as fine straight lines and confirmed by magnetic particle inspection, which reveals their length and depth pattern more clearly than the unaided eye.

(d) Bursting (Burst / Rolling Burst)

Bursting refers to a gross rupture or splitting-open of the bar surface (and often into the body of the bar) that occurs when the steel's ductility at rolling temperature is insufficient to accommodate the deformation being imposed — commonly due to rolling at too low a temperature, excessive reduction taken in a single pass, or the presence of internal defects (such as porosity, non-metallic inclusion clusters, or centre segregation) inherited from the cast billet that act as internal stress concentrators. A burst typically appears as a ragged, torn opening, often transverse or at an angle to the rolling direction, sometimes exposing an internally porous or coarse-grained fracture surface. Because bursting compromises the structural integrity of the bar at that location, affected sections are always rejected and cut out; recurring bursting is investigated as a process issue relating to furnace soaking temperature, pass schedule, or incoming billet quality rather than treated as an isolated grinding-repair defect.

(e) Scratch

Scratches are shallow, mechanically-induced linear marks on the bar surface caused by contact with fixed guides, worn roller tables, misaligned guide boxes, or handling equipment as the bar travels through the mill, cooling bed, or finishing lines. Unlike laps or seams, scratches are of purely mechanical origin (not rolled-in from the billet) and generally do not involve folded metal or a pre-existing crack; they appear as bright, shallow grooves or scuff marks, often with a consistent depth and a somewhat polished appearance from the sliding contact. Minor scratches within depth tolerance are generally acceptable, while deeper scratches that reduce the effective section beyond permissible limits are ground out; recurring scratching is corrected by inspecting and realigning or renewing worn guides and roller table components.

(f) Rib Defects

For ribbed (deformed/TMT) bars, the transverse and longitudinal ribs are formed by the finishing pass groove pattern, and "rib defects" describe deviations in this rib geometry — including uneven, incomplete, or asymmetric rib height, rib spacing, or rib angle, as well as excess flash or fin metal along the bar's longitudinal rib line where the roll groove parting line does not close cleanly. Common causes include worn or damaged finishing roll grooves, incorrect roll gap setting, guide misalignment feeding the bar off-centre into the finishing pass, or insufficient/excess stock size approaching the finishing stand. Rib defects are checked during dimensional inspection using rib-profile gauges against the applicable bar standard, since rib geometry directly affects bond strength with concrete; out-of-tolerance rib pattern is corrected by finishing-stand roll change or realignment rather than by reconditioning the already-rolled bar.

(g) Rollout (Rolled-in Defects)

"Rollout" refers to any pre-existing billet surface or near-surface defect — such as a billet-surface crack, cold shut, scab, or trapped scale patch — that is progressively elongated and "rolled out" along the bar's length as the billet passes through the reduction sequence, rather than being closed up or welded shut by the rolling pressure. A rolled-out defect typically appears as an elongated, sometimes intermittent surface irregularity that is much longer and shallower in the finished bar than the original billet defect, making its origin sometimes difficult to identify without cross-sectional metallography. Preventing rollout defects depends primarily on effective billet inspection and conditioning (grinding) before charging, since once such a defect enters the roughing mill, it cannot be reliably removed by subsequent rolling; in the finished bar, rolled-out defects are managed the same way as laps/seams — assessed for depth, and ground out or the section rejected if beyond tolerance.

(h) Pinhole / Pin Crack

Pinholes are minute, fine surface openings - often only a fraction of a millimetre in diameter, that originate from small gas pores or subsurface blowholes present just beneath the billet surface at the time of casting; when this region is rolled down, the pore is drawn out and can break through to the surface as a tiny pit or pinhole, or remain as a very fine "pin crack" a short, hairline surface crack, sometimes clustered in groups. These defects are frequently associated with entrapped gas from moisture or from casting powder breakdown products (see below) at the solidification front during continuous casting. Individually small, pinholes and pin cracks can nonetheless act as stress concentrators and, if numerous or clustered, can significantly affect fatigue life and surface integrity; they are best detected by magnetic particle inspection, which reveals clustering patterns not always obvious to the unaided eye, and are addressed through improved billet surface conditioning and, at the casting stage, better mould powder and moisture control.

(i) Inclusion / Entrapment Defects (Casting Powder and Moisture Related)

A further class of defects originates not from the rolling operation itself but from material entrapped in the steel during continuous casting, which is subsequently exposed or elongated during rolling. Mould flux (casting powder) is added at the top of the continuous casting mould to lubricate the strand, absorb inclusions, and provide thermal insulation; however, if flux is entrained into the solidifying steel — due to excessive mould level fluctuation, improper flux viscosity/melting characteristics, or turbulent flow in the mould — it becomes trapped as a non-metallic inclusion just beneath the cast surface. When the billet is subsequently rolled, these entrapped flux inclusions elongate along the rolling direction and can surface as glassy, dark-coloured streaks or open up as a shallow surface defect similar in appearance to a seam or lap, but distinguishable on closer or metallurgical examination by the presence of oxide/silicate inclusion material rather than a simple fold of parent steel. Separately, residual moisture — whether present in the mould flux itself, in refractories, or absorbed into billets during outdoor storage in wet conditions — can flash to steam when the billet is subsequently reheated, generating internal gas pressure that produces subsurface blowholes or, in more severe cases, contributes to the bursting defects described above. Control of these defects is primarily achieved upstream, through proper mould flux selection and drying, stable casting practice to minimise mould level fluctuation, and covered/dry billet storage to avoid moisture pickup prior to reheating.

9. Positive Material Identification (PMI)

9.1 Overview and Purpose

Positive Material Identification (PMI) is a non-destructive testing technique used to verify the chemical composition of a steel product and confirm that it matches the declared grade and specification. In a rolling mill environment, PMI is performed on finished rolled bars and sections to provide documented assurance that the material delivered to the customer is exactly what was ordered — preventing grade mix-ups, mislabelling, or inadvertent use of incorrect heats in critical applications. PMI is especially critical for alloy steel grades, stainless steels, and high-strength structural steels where even minor deviations in key alloying elements can significantly alter mechanical performance, weldability, and corrosion resistance.

9.2 Principle of Operation

9.2.1 X-Ray Fluorescence (XRF) Method

The most widely used PMI technique in rolling mill quality assurance is Energy-Dispersive X-Ray Fluorescence (ED-XRF). The instrument directs a primary X-ray beam from a miniature X-ray tube onto the steel surface. This beam excites the atoms in the material, causing inner-shell electrons to be ejected. As outer-shell electrons fill these vacancies, each element emits characteristic fluorescent X-rays at energies unique to that element.

A solid-state detector — typically a silicon drift detector (SDD) — collects these fluorescent X-rays and produces an energy spectrum. The instrument's software analyses peak positions (which identify elements) and peak intensities (which quantify concentrations), delivering a full elemental composition result in 20 to 60 seconds without any sample preparation or surface damage.

> *Key Detectable Elements via XRF: Fe, C*, Mn, Si, P, S*, Cr, Mo, Ni, Cu, V, Nb, Ti, Al, Co, W, B**
>
> ** Carbon, Sulphur, and Boron are light elements with limited XRF sensitivity. OES is preferred for these elements.*

9.2.2 Optical Emission Spectrometry (OES) — Laboratory Verification

For grade verification requiring accurate carbon, sulphur, and boron measurement, Optical Emission Spectrometry (OES) is performed in the laboratory. A high-energy electrical spark or arc is applied to a prepared flat sample surface, exciting atoms to emit characteristic light at wavelengths unique to each element. A spectrometer separates and measures these wavelengths simultaneously, providing precise quantitative analysis of all major and trace alloying elements within seconds. OES is the primary laboratory instrument for heat certification and is used to confirm PMI results when borderline compositions are detected by XRF in the field.

9.3 Testing Process — Step by Step

Step 1: Select sample bars from the finished lot per the applicable sampling plan (typically one bar per heat per size).

Step 2: Clean the surface of the test area using a wire brush or grinding disc to remove mill scale, rust, or surface contamination that could affect the reading.

Step 3: Place the XRF analyser gun firmly against the prepared surface, ensuring full aperture coverage on the steel.

Step 4: Activate the instrument and allow the measurement cycle (30–60 seconds). The instrument displays elemental percentages in real time.

Step 5: Compare the measured composition against the target grade chemistry limits as per the applicable standard or customer specification.

Step 6: Record results in the PMI test register or export to the quality management system. Apply PASS/FAIL disposition.

Step 7: For any FAIL or borderline result, segregate the material and initiate laboratory OES verification before making a final disposition.

9.4 Equipment and Calibration

PMI analysers must be calibrated against certified reference standards covering the grade range being tested. Calibration verification is performed at the start of each shift using a certified calibration check sample. The instrument calibration certificate and last verification record must be traceable to national standards.

9.5 Acceptance Criteria

Parameter	Requirement
Carbon (C)	Per grade specification (e.g., ≤ 0.30% for TMT Fe-500)
Manganese (Mn)	Per grade specification (e.g., ≤ 1.60%)
Chromium (Cr)	As specified; critical for alloy grades
Molybdenum (Mo)	As specified; tolerance typically ± 0.02%
Nickel (Ni)	As specified
Carbon Equivalent (CE)	As per weldability requirement (e.g., CE ≤ 0.42% for structural grades)

Figure 9: **PMI Machine**

10. Magnetic Particle Inspection (MPI)

10.1 Overview and Purpose

Magnetic Particle Inspection (MPI) is a non-destructive surface and near-surface defect detection technique applicable to ferromagnetic materials such as carbon and alloy steels. In post-rolling inspection, MPI is used to detect surface-breaking and near-surface discontinuities including seams, laps, cracks, folds, and linear surface flaws that may have originated during rolling, straightening, or cooling — including the lap, seam, and pin-crack defects described in Section 8.3. MPI is highly sensitive to surface-breaking and shallow sub-surface defects, making it a preferred method for bars and billets where surface integrity is critical for machining, heat treatment, or end-use structural applications.

10.2 Principle of Operation

When a ferromagnetic material is magnetised, magnetic flux lines flow through the material parallel to its length. If a surface or near-surface discontinuity is present, it disrupts the path of the magnetic flux lines, causing them to leak out of the material surface above and around the defect — a phenomenon known as a flux leakage field. Fine ferromagnetic particles — either dry powder or suspended in a carrier liquid — are applied to the magnetised surface. These particles migrate to and concentrate at the flux leakage points, forming a visible indication that outlines the shape and extent of the underlying discontinuity. Under appropriate lighting (white light for coloured particles; ultraviolet/UV light for fluorescent particles), these indications are clearly visible to the inspector.

Detection Sensitivity: Surface-breaking defects as narrow as 0.001 mm width are detectable.

Near-surface depth: MPI reliably detects discontinuities up to approximately 3–6 mm below the surface, depending on field strength and particle type.

Limitation: MPI is not applicable to non-ferromagnetic materials (e.g. austenitic stainless steels).

10.3 Magnetisation Techniques

10.3.1 Longitudinal Magnetisation (Coil / Solenoid Method)

A coil wound around the bar passes an AC or DC current, inducing a longitudinal magnetic field along the bar's axis. This technique is most effective for detecting transverse (circumferential) defects such as transverse cracks and seams perpendicular to the bar length.

10.3.2 Circular Magnetisation (Direct Contact / Prod Method)

Current is passed directly through the bar between contact electrodes, creating a circular (circumferential) magnetic field around the bar cross-section. This method is best suited for detecting longitudinal defects — seams, laps, and axial cracks — running parallel to the bar axis.

10.3.3 Combined Magnetisation — Multi-Directional

Modern MPI systems for bar inspection apply simultaneous longitudinal and circular fields, ensuring detection of discontinuities in all orientations in a single pass. Automated MPI lines for rolled bars commonly use this approach to maximise throughput and detection reliability.

10.4 Wet Fluorescent MPI Process — Step by Step

Step 1: De-scale and clean the bar surface. Fluorescent MPI requires a clean, oil-free surface for reliable particle migration.

Step 2: Set up the magnetisation system. For bar inspection, pass the bar through a coil or position between contact heads at the correct spacing.

Step 3: Apply the fluorescent magnetic particle suspension (bath) uniformly over the test area while the magnetising current is active.

Step 4: Apply the magnetising current (typically 800–3000 A depending on bar diameter) for the required dwell time (2–5 seconds).

Step 5: Immediately examine the surface under UV-A light (365 nm) in a darkened inspection booth. Fluorescent indications appear as bright yellow-green lines against a dark background.

Step 6: Photograph any relevant indications and measure their length and orientation.

Step 7: Classify indications per acceptance criteria. Relevant linear indications exceeding the specified limit are cause for rejection.

Step 8: Demagnetise the bar after inspection using a decreasing AC demagnetisation coil.

Figure 10: **Visible defects via MPI**

10.5 Acceptance Criteria

Indication Type	Typical Acceptance Limit
Linear surface indications	No linear indication > 3 mm length (or per spec)
Transverse cracks	Not permitted — zero tolerance
Seams and laps	Not permitted beyond agreed surface depth limit
Non-relevant indications	Documented; no action required

Figure 11: **MPI Workstation**

11. Automated Ultrasonic Testing (Auto UT)

11.1 Overview and Purpose

Automated Ultrasonic Testing (Auto UT) is a volumetric non-destructive testing method that uses high-frequency sound waves to detect internal discontinuities within the steel bar. Unlike MPI — which is limited to surface and near-surface flaws — Auto UT provides full volumetric coverage of the bar cross-section, detecting internal defects such as pipe, porosity, segregation, bursts, cracks, inclusions, and laminations that are invisible from the surface. In a rolling mill, Auto UT is performed on finished bars either inline (immediately after the rolling and cooling line) or offline on a dedicated UT inspection bench, fully automated for 100% volumetric inspection of production output at production speeds.

11.2 Principle of Operation

Ultrasonic testing uses piezoelectric transducers (probes) to generate high-frequency sound pulses — typically 2 to 10 MHz for steel bar inspection — and transmit them into the material. The sound waves travel through the steel and are reflected when they encounter a boundary between materials of different acoustic impedance — including the far wall of the bar, or any internal discontinuity such as a crack, void, or inclusion. The reflected echoes are received by the same transducer (pulse-echo method) or a separate receiver probe. The time elapsed between transmission and reception of the echo (time-of-flight) is used to calculate the depth of the reflector, and the amplitude of the echo signal is compared against a calibrated reference level to classify the size and significance of the indication.

Frequency Range: 2 – 10 MHz (higher frequency = better resolution but lower penetration depth)
Coupling Medium: Water immersion (most common for automated bar testing) or gel couplant
Coverage: 100% volumetric inspection of bar cross-section per production piece
Detection Threshold: Flat-bottomed hole (FBH) equivalent reflectors as small as 0.5 mm diameter detectable

11.3 Auto UT System Configuration

11.3.1 Immersion Testing Setup

In the most common configuration for round bar Auto UT, the bar passes through a water-filled test tank at controlled speed. Multiple fixed-position transducer arrays — angled to provide helical scan coverage — simultaneously transmit pulses through the water couplant into the rotating or axially moving bar. The complete circumference and cross-section are covered in a helical scan pattern ensuring no volume is missed.

11.3.2 Phased Array Ultrasonic Testing (PAUT)

Advanced Auto UT lines use Phased Array probes containing multiple piezoelectric elements that can be individually time-delayed. This allows electronic beam steering and focusing, providing higher resolution, better near-surface detection, and greater flexibility in coverage angles compared to single-element probes — all without mechanical probe movement.

11.3.3 Time-of-Flight Diffraction (TOFD)

TOFD is a complementary technique used for accurate sizing of internal defects detected during standard UT. It uses diffracted signals from defect tips rather than reflected signals, providing more accurate depth measurement and length sizing of planar defects such as internal cracks and laminations.

11.4 Auto UT Inspection Process — Step by Step

Step 1: System calibration: Set the UT instrument gain, pulse repetition frequency, and gate settings using a calibration test block (reference standard) containing flat-bottomed holes or notches of known dimensions. Calibration must be performed at the start of each shift and after any equipment change.

Step 2: Bar entry: The finished bar enters the auto UT station via roller conveyors. Surface condition and straightness are checked — excessive bend or surface scale can degrade coupling and mask signals.

Step 3: Helical scan: The bar passes through the immersion tank or between water-jet coupled probes at a controlled feed rate. Multiple probe channels simultaneously inspect the bar from different angles to ensure full volumetric coverage.

Step 4: Real-time signal acquisition: The UT electronics system continuously acquires A-scan (waveform), B-scan (cross-section image), and C-scan (plan-view map) data for each bar, building a complete volumetric record.

Step 5: Automated alarm: When a reflected signal exceeds the pre-set alarm threshold (referenced to the calibration reflector), the system triggers a location-tagged alarm and marks the defect position on the bar.

Step 6: Bar sorting: At the exit of the UT station, bars receiving an alarm are automatically diverted to a reject or hold conveyor for further investigation. Bars with no relevant indications pass to the accepted product stream.

Step 7: Data recording: All inspection data — including bar ID, scan map, alarm locations, signal amplitudes, and calibration records — are automatically stored in the quality data management system for traceability.

11.5 Acceptance Criteria

Defect Category	Acceptance Limit (Typical)
Single point reflector	Echo amplitude < 50% DAC (Distance Amplitude Curve)
Clustered indications	Not permitted in critical zones
Continuous linear indication	Not permitted — rejection criterion
Pipe / Axial void	Zero tolerance — automatic rejection
Near-surface zone	Checked separately with shallow-angle probe; FBH ≥ 1.0 mm not accepted

Figure 12: **Auto UT Machine**

12. Macro Examination

12.1 Overview and Purpose

Macro examination is a metallurgical inspection technique in which a transverse or longitudinal section of a rolled bar is cut, prepared, and etched with an appropriate acid reagent to reveal the macrostructure of the steel at low magnification (typically the naked eye or up to 10×). The technique exposes the solidification pattern, segregation zones, flow lines, internal defects, and the overall soundness of the steel as it solidified in the continuous caster and was subsequently deformed during rolling. Macro examination is a routine quality check performed on samples from each heat — typically one bar end section per heat per product size — and is essential for validating the quality of the continuously cast and rolled product before release.

12.2 Specimen Preparation

Step 1 — Cutting: A transverse slice of approximately 15–25 mm thickness is cut from the bar end (cropped end or a dedicated test bar sample) using a band saw, abrasive cut-off wheel, or power hacksaw. Cuts must be perpendicular to the bar axis.

Step 2 — Surface Preparation: One face of the slice is ground flat on a surface grinder or belt grinder to a smooth finish (typically 180–400 grit). Surface flatness is critical for uniform etching.

Step 3 — Cleaning: The prepared surface is thoroughly cleaned with acetone or industrial alcohol to remove grinding debris, oil, and moisture. Any contamination will cause uneven or false etching.

Step 4 — Etching: The cleaned surface is immersed in hot hydrochloric acid solution (50% HCl in water at 65–80°C) for 15 to 30 minutes (Baumann etch or Fry's etch is also used for specific features). The acid preferentially attacks segregated regions, grain boundaries, and defect areas, revealing the macrostructure through differential dissolution.

Step 5 — Neutralisation and Drying: The etched specimen is removed from the acid bath, immediately rinsed with hot water, neutralised in a sodium bicarbonate solution, rinsed again, and dried (compressed air or oven). Ethanol rinse before drying improves surface appearance.

Step 6 — Examination: The etched surface is examined visually and photographed under white light. A rating is assigned based on applicable standards.

12.3 Features Revealed by Macro Etching

- Central segregation (primary and secondary) — enrichment of carbon, sulfur, and phosphorus at the solidification centre
- Centre porosity and pipe — shrinkage voids at the centreline from solidification contraction
- Sub-surface porosity — gas or shrinkage voids in the subsurface zone
- Dendritic solidification structure — the pattern of primary solidification
- Flow lines — deformation flow pattern from rolling, confirming material continuity
- Cracks — internal hot tears, hydrogen flakes, or rolling cracks
- Segregation bands — alternating high and low alloy bands from rolling of segregated cast structure
- White spots / flaking — hydrogen embrittlement cracks (critical in alloy and tool steels)

12.4 Rating and Acceptance

Macro examination results are rated using standard rating charts from applicable standards, most widely SEP 1572 (Stahl-Eisen-Prüfblatt), ASTM E381, and EN ISO 4969.

Feature	Rating Scale	Typical Acceptance Limit
Centre segregation	0 to 4 (SEP 1572 / Mannesmann scale)	≤ 2.0 for standard grades; ≤ 1.5 for critical grades
Centre porosity	0 to 4	≤ 1.5
Sub-surface porosity	0 to 4	≤ 1.0
Cracks (internal)	Present / Absent	Absent — zero tolerance
Pipe	Present / Absent	Absent — zero tolerance
White spots	Present / Absent	Absent — zero tolerance

Figure 13: **Macro Examination Specimen**

13. Laboratory Testing

The Metallurgical Laboratory performs a range of mechanical and metallurgical tests on samples drawn from finished rolled products. These tests provide quantitative validation of material properties against specification requirements and are the basis for the material test certificate (MTC) issued with each product batch.

13.1 Hardness Testing

Hardness testing measures a material's resistance to localised plastic deformation induced by a standardised indenter under a defined load. In rolled steel products, hardness testing serves multiple purposes: verifying the as-rolled mechanical condition, confirming heat treatment effectiveness (for treated grades), checking for decarburisation at the surface, and providing a rapid screening check correlated to tensile strength.

13.1.1 Brinell Hardness Test (HB)

A hardened steel or tungsten carbide ball (10 mm diameter) is pressed into the polished test surface under a load of 3000 kgf for 10–15 seconds. After load removal, the diameter of the residual circular impression is measured using an optical microscope or digital measurement system. The Brinell Hardness Number (HBW) is calculated as the applied load divided by the curved surface area of the indentation. Brinell hardness is the preferred method for rolled bars and sections due to the large indenter averaging effects over a representative area.

13.1.2 Rockwell Hardness Test (HR)

A diamond cone (Brale indenter) or hardened steel ball is applied under a minor load followed by a major load. The hardness number is derived from the net increase in indentation depth due to the major load, read directly from the instrument scale. Rockwell B (HRB) and Rockwell C (HRC) are the most common scales for steel, providing a quick, direct-reading result used for rapid quality screening in production environments.

13.1.3 Hardness Testing Process

Step 1: Cut the sample transversely and grind/polish the test face to a metallographic finish (at minimum 600 grit, preferably polished to 1 µm for Vickers).

Step 2: Place the specimen flat on the testing machine anvil. Ensure surface is clean and parallel.

Step 3: Apply the indenter under the specified load for the required dwell time. Ensure vibration-free conditions during loading.

Step 4: Measure the indentation dimensions (Brinell and Vickers: optical measurement; Rockwell: direct digital readout).

Step 5: Calculate and record the hardness number. For cross-section surveys, repeat at specified intervals from surface to core.

Step 6: Compare against acceptance criteria in the applicable product standard.

Figure 14: (a) Brinell Hardness Tester (b) Rockwell Hardness Tester

Test Method	Indenter	Load Range	Primary Application
Brinell (HBW)	10 mm WC ball	3000 kgf	Bars, sections — bulk hardness
Vickers (HV)	Diamond pyramid	1 – 30 kgf	Cross-section traverses, microhardness
Rockwell (HRB/HRC)	Ball / Diamond cone	60–150 kgf	Rapid production screening

13.2 Hardenability Testing - Jominy End-Quench Test

Hardenability is the property of a steel that describes how deeply and uniformly it can be hardened by quenching from the austenitising temperature. It is distinct from hardness (the absolute level of hardness achievable) — hardenability relates to the depth of the hardened zone. Hardenability is a critical parameter for alloy steel bars intended for through-hardened components such as shafts, gears, axles, and fasteners. The standard method for measuring hardenability is the Jominy End-Quench Test, standardised in ASTM A255, IS 3848, and EN ISO 642.

13.2.1 Test Procedure

Step 1 — Sample Preparation: Machine a standardised Jominy test bar: 25 mm diameter × 100 mm long with a flange at the top. The specimen must be machined from the actual product or from the same heat under test.

Step 2 — Austenitising: Heat the Jominy specimen to the specified austenitising temperature (typically 860°C for carbon and alloy steels; temperature per applicable standard for the grade) and hold for 30 minutes to achieve complete and uniform austenitisation.

Step 3 — End-Quench: Transfer the specimen immediately to the Jominy end-quench fixture within 5 seconds. A controlled column of water at 24°C impinges on one flat end face of the specimen from a 12.5 mm nozzle at a specified flow rate for 10 minutes. The free end is quenched rapidly; the remaining length cools progressively more slowly away from the quenched end.

Step 4 — Flat Grinding: After quenching, grind two diametrically opposite flat faces along the full length of the specimen to a depth of 0.38 mm to remove the decarburised surface layer and expose fresh steel for hardness measurement.

Step 5 — Hardness Traverse: Measure Rockwell C hardness at standardised distances from the quenched end: at 1.5 mm, then every 1.5 mm up to 15 mm, then every 3 mm up to 60 mm, per ASTM A255.

Step 6 — Hardenability Band Plot: Plot the HRC values against distance from quenched end to produce the Jominy hardenability curve. Compare against the specified hardenability band (H-band) for the grade.

> *Interpretation: A steep drop in hardness with distance indicates low hardenability (shallow-hardening steel).*
>
> *A flat curve indicates high hardenability (deep-hardening steel).*
>
> *The J-distance at which HRC drops to 50% of the maximum value is often used as a reference hardenability index.*

Figure 15: **Jominy End Quench Tester**

13.3 Non-Metallic Inclusion Rating

Non-metallic inclusions are phases present in steel that are not part of the base metallic matrix — they originate from deoxidation products, refractory erosion, slag entrapment, and precipitation of sulphides and oxides during solidification and cooling. While some inclusions are unavoidable in steelmaking, their type, size, morphology, and distribution significantly affect the steel's fatigue strength, toughness, ductility, machinability, and surface quality of the finished product. Inclusion rating is a mandatory quality test for alloy steels, bearing steels, spring steels, cold heading steels, and any application where cleanliness is specified, and is also conducted as a statistical process control measure for carbon steel grades.

13.3.1 Inclusion Types - Classification

Type	Designation	Composition	Morphology
Sulphides	Type A	MnS, FeS	Grey, elongated stringers — plastic deformation during rolling
Alumina	Type B	Al_2O_3 (aluminate)	Angular, clustered — brittle, aligned in rolling direction
Silicates	Type C	SiO_2, complex silicates	Glassy, elongated — form chains during rolling
Globular Oxides	Type D	Complex CaO-Al_2O_3-SiO_2	Spherical, randomly distributed — not deformed by rolling
Single Oxides	Type DS	Globular but large single oxides	Spherical, > 13 µm — treated separately from Type D

13.3.2 Sample Preparation for Inclusion Rating

A longitudinal metallographic section is cut from the mid-radius position of the bar (the 1/2-radius location is the most representative area for rolled bar cleanliness assessment). The section must be cut parallel to the rolling direction, as inclusion morphology and distribution are strongly direction-dependent. The sample is mounted in bakelite or epoxy, ground through successive silicon carbide papers (120 → 240 → 400 → 600 → 800 → 1200 grit), and polished with 3 µm followed by 1 µm diamond suspension on polishing cloth until a mirror finish is achieved. The sample is cleaned ultrasonically in ethanol and dried before examination.

13.3.3 Rating Method - Microscopic Examination

Inclusion rating is performed at 100× magnification on an optical microscope equipped with a calibrated eyepiece graticule. The entire prepared surface area — or a minimum defined area per the applicable standard — is systematically scanned, and each field of view is compared against standard reference chart images representing increasing severity levels of each inclusion type and thickness category. Two sub-categories are assessed for each inclusion type: thin series (suffix e, finer inclusions below a specified width threshold) and heavy series (suffix d, coarser, more severe inclusions above the width threshold). The most severe field rating observed across the scanned area is reported as the inclusion rating for each type and series combination, on a scale ranging from 0 (absent) to 3 or 4 (very heavy) depending on the standard used.

13.3.4 Standards for Inclusion Rating

Standard	Method	Notes
ASTM E45	Chart method (Methods A, B, C, D)	Most widely used globally; four chart methods available
EN 10247	Chart method (SEP 1570 based)	European standard; comparable to DIN 50602
DIN 50602	Chart method M (worst field)	German standard; commonly referenced for bearing steels
IS 4163	Chart comparison method	Indian standard for inclusion rating of steel
SEP 1570	Chart method	Stahl-Eisen-Prüfblatt; frequently specified in Europe

13.3.5 Acceptance Criteria

Grade / Application	Max Type A	Max Type B	Max Type C	Max Type D
Standard structural steel	2.5 (thin)	2.0 (thin)	1.5 (thin)	2.0
Engineering alloy steel	2.0 (thin)	1.5 (thin)	1.0 (thin)	1.5
Bearing steel (52100 / 100Cr6)	1.0	1.0	0.5	1.0
Spring steel / Cold heading	1.5	1.0	0.5	1.0

14. Testing Programme Summary

Test	Type	What It Detects	Standard Reference
PMI (XRF / OES)	Chemical / NDT	Grade verification; elemental composition	ASTM E1479; IS 228
MPI	Surface NDT	Surface and near-surface cracks, seams, laps	ASTM E709; EN ISO 9934
Auto UT	Volumetric NDT	Internal defects: pipe, cracks, voids, inclusions	ASTM A388; EN 10308
Macro Examination	Metallurgical	Segregation, porosity, pipe, cracks, flow lines	ASTM E381; EN ISO 4969
Hardness Testing	Mechanical Lab	Surface / bulk hardness; strength correlation	ASTM E10/E18; IS 1500
Hardenability (Jominy)	Mechanical Lab	Depth of hardening; response to heat treatment	ASTM A255; EN ISO 642
Inclusion Rating	Metallurgical Lab	Non-metallic cleanliness; inclusion type and severity	ASTM E45; EN 10247

PMI confirms that the correct grade was produced and no inadvertent material mix has occurred. MPI ensures that no surface or near-surface cracking is present that could compromise the product in service or during downstream processing. Auto UT provides confidence in the internal soundness of the bar throughout its full volume. Macro examination reveals the solidification and rolling quality of each heat. Laboratory testing of hardness, hardenability, and inclusion rating completes the picture by providing quantitative mechanical and metallurgical data that underpin the material test certificate.

15. Defect Reconditioning

15.1 Defect Identification

Defects located during visual inspection, MPI, Auto UT, or macro examination are precisely located along the bar length and their type, extent, and depth are recorded, forming the basis for the disposition decision — accept, recondition, or reject — applied to that section of bar.

15.2 Chalk Marking Procedure

Once a defect is confirmed, its location and boundary are marked directly on the bar surface using chalk or paint markers, clearly delineating the area to be worked on so that grinding personnel remove only the affected region and the minimum necessary material.

15.3 Grinding Operation

The marked area is locally ground, typically using a hand-held pneumatic or electric grinder, to remove the defect completely while blending the ground area smoothly into the surrounding surface to avoid creating a new stress concentration. Grinding depth is monitored against the applicable minimum remaining wall/diameter tolerance for the product, since excessive grinding can itself render the bar undersized and therefore non-conforming.

Figure 16: **Grinding**

15.4 Reinspection after Grinding

After grinding, the reconditioned area is reinspected — visually, and by MPI where the original defect was crack-like — to confirm complete removal of the defect and to verify that the remaining cross-section and surface condition meet specification before the bar is passed forward to final inspection.

Figure 17: **Reinspection after Grinding**

16. Quality Assurance, Process Control, and Final Inspection

16.1 In-Process Control Parameters

Consistent product quality in rolling mill operations depends on rigorous process control across all production stages. The following parameters are continuously monitored and controlled:

- Billet temperature at furnace discharge (infrared pyrometry)
- Descaling pressure and water flow confirmation
- Roll gap and roll force at each stand (load cell feedback)
- Rolling speed synchronisation between successive stands
- Finishing temperature verification (optical pyrometer)
- Cooling bed residence time and cooling rate
- Final dimensional checks (laser gauge, manual sampling)
- Mechanical property testing (tensile, yield, bend, elongation)

Non-conforming material identified during in-process or final inspection is segregated, tagged, and subjected to disposition review before any release or rework decision. Full heat number traceability from ladle to finished bundle ensures accountability at every processing step.

16.2 Acceptance Criteria and Product Approval

Final acceptance is based on the complete set of inspection and testing results — visual surface condition, dimensional measurements, PMI grade confirmation, MPI and Auto UT results, macro examination rating, and laboratory hardness/hardenability/inclusion data — assessed against the applicable national or customer product standard, with formal sign-off recorded against the batch/heat number. Bars and bundles meeting all acceptance criteria are formally approved for dispatch by the quality assurance department, with an inspection/test certificate generated summarising the batch's chemical composition, mechanical properties, and inspection results.

16.3 Packing and Dispatch

Approved bundles are prepared for transport with appropriate end-capping, banding, and, where required, weather-protective wrapping, and are tagged with all identification details required by the customer purchase order and applicable standard. Packed bundles are loaded onto transport vehicles or railway wagons according to the dispatch schedule, accompanied by the test certificate, delivery challan, and any other documentation required for the consignment, completing the traceable chain from billet receipt through to customer delivery.

17. Conclusion

The rolling mill process is a precisely sequenced thermal and mechanical operation that transforms continuously cast steel billets into dimensionally accurate, structurally sound, and surface-quality finished products. Each stage — from reheating and descaling through roughing, intermediate, and finishing rolling to post-rolling treatment — is interdependent, and performance at any stage directly influences final product quality. Optimal mill operation requires disciplined management of billet temperature profiles, rolling schedules, roll pass geometry, guide alignment, cooling practice, and dimensional verification.

The post-rolling testing programme described in this report constitutes a comprehensive, multi-layered quality assurance system that validates steel product integrity from chemical composition through to internal cleanliness and mechanical suitability. Each test method addresses a specific failure mode or quality dimension that cannot be adequately covered by the other techniques alone, and together these methods ensure that every batch of rolled steel product released from the mill meets the exacting quality standards required by construction, infrastructure, automotive, energy, and precision engineering applications.

The rolled steel products produced and certified through this combined process — including TMT reinforcement bars, wire rods, rounds, flats, and structural sections — serve critical structural and engineering applications across construction, infrastructure, automotive, and general manufacturing industries, backed by full traceability and documented quality assurance from heat to finished, dispatched bundle

— End of Report —